ALLEN PRASAD VARGHESE

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Skills Summary

- Developed a high-impact polymer extrusion die design algorithm, reducing design cycle time by 98%.
- Strong capabilities in technical research, systems engineering, and project management.
- Proficient in multiple programming languages to automate and enhance engineering workflows.
- Extensive research experience in fluid systems like bipropellant liquid rocket fundamentals and ability to collaborate with subject matter experts to maximize engine performance.
- Strong communication abilities with ability to clearly articulate complex engineering concepts.
- I am adept at reading interpersonal dynamics and responding empathetically.

Key Industry Skills:

- Six Sigma Green Belt Certified
- Computer Aided Engineering (CAE) – ANSYS Fluent, DualSPHysics, Thermal Desktop, and Converge.
- Computer Aided Design (CAD) – SolidWorks, CREO, and Fusion 360.
- Programming – UDF DevOps, MATLAB, VBA, C#, C++, Fortran, JavaScript, Python, and OpenMPI.
- HPC Linux Clusters – Kong, XSEDE Bridges, Stheno, Wulver GPU, SwRI HPCs and Azure HPC.

Work Experience (2018-2026)

1. **Mechanical and Aerospace AI Contractor,** *May 2026 – Present*
   Mercor Intelligence
   Salary: $80 hourly
   Hours per week: 40

   - Developed training data for large language models by formulating coding problems that models could not resolve and subsequently documenting the correct solutions.
   - Maintained high accuracy and throughput in large-scale annotation workflows, contributing to reliable training data for AI model fine-tuning.

2. **Propulsion Engineer,** August 2024 – January 2026
   Southwest Research Institute (SwRI)
   Liquid Propulsion Group
   Hours per week: 40
   Supervisor: Nathan Andrews

   - Designed and came up with thruster profiles that target favorable fluid orientations during micro-gravity operations at low-Earth orbit and deep space near/on the moon.
   - Participated in and led testing campaigns to go over experimental matrices to understand fluid behaviors pertaining to the aerospace sector mainly sloshing.
   - Conduct Computational Fluid Dynamics simulations for various aerospace clients towards predicting ullage behavior for NTO, MONx and MMH tanks.

3. **Research Assistant,** August 2018 – August 2024
New Jersey Institute of Technology (NJIT)

CFD Hydrodynamic Analysis Experience
Funded by: National Science Foundation (NSF)
Supervisor: Dr. Angelo Tafuni

- Spearheaded computational efforts using expertise in DualSPHysics to establish novel hydrodynamics for vibrating beams submerged underwater near the free surface.
- Code development as well as prowess in MATLAB data analysis as evidenced by developing an algorithm to extract relevant hydrodynamic coefficients from fluid forces.

CFD Cryogenic Multiphase Flow Analysis Experience
Funded by: National Aeronautics and Space Administration (NASA)
Supervisor: Dr. Jason W. Hartwig, Dr. Angelo Tafuni & Dr. Samuel C. Lieber

- Cross-functional collaborator with acumen in design of cryogenic propellant management systems that delivered 92% expulsion efficiency under reduced gravity conditions.
- Complex CFD model development with extensive knowledge of cryogenic propellants, accurately describing flow in microgravity for long duration space mission.
- Demonstrated strong project management and technical research capability, leading to 2 publications in journal of AIAA, and Journal of Spacecraft and Rockets.

Polymer Process Automation Experience
Funded by: New Jersey Precision Technologies (NJPT)
Supervisor: Robert Tarantino, Dr. Samuel C. Lieber & Dr. Angelo Tafuni

- Process automation mastery with experience in pioneering parametric design automation using SolidWorks API to accelerate polymer extrusion die design by 98%.
- Expert system developer with ability to construct self-learning knowledge databases drawing automated inferences from prior design cases to tackle the growing skills gap.
- Knowledge transfer facilitator with exposure in interviewing interdisciplinary engineers, to develop 4 robust standard working protocols based on gathered insights and bottlenecks across the organization.
- The assessment toolkit involved a parametric design automation script using SolidWorks API to accelerate polymer extrusion die design, and a finite element (FE) analysis model using ANSYS Polyflow.

Education

Doctor of Philosophy in Mechanical Engineering (PhD), *5 Years* **Graduated**
New Jersey Institute of Technology

Master of Science in Mechanical Engineering (MS), *2 Years* **Graduated**
New Jersey Institute of Technology

Master of Engineering in Mechanical Engineering (MEng), *4 Years* **Graduated**
Lancaster University